Software Engineering Requirements

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Software Engineering

- Software engineering is a discipline of computer science that is concerned with the software development process: design, development, testing, and maintenance of software applications
- ACM definition:
  - "Software engineering spans the entire software lifecycle - it involves creating high-quality, reliable programs in a systematic, controlled, and efficient manner using formal methods for specification, evaluation, analysis and design, implementation, testing and maintenance."
    - https://ccecc.acm.org/guidance/software-engineering
  - Software Engineering was created as a program of study in 2005

Principles of Software Engineering (1)

- Must be viewed as a discipline with stronger ties to CS than it has to other engineering fields
- Must share common characteristics with other engineering disciplines, including:
  - quantitative measurement
  - structured decision making
  - effective use of tools
  - reusable software components.
- Must apply engineering methods and practices to the development of software, with special emphasis on the development of large software systems to create extensible software

Principles of Software Engineering (2)

- Must integrate the principles of mathematics and CS with engineering methodologies to create extensible software
- Must include secure code standards and quality control concepts of manufacturing process design
- Must emphasize communication skills, teamwork skills, and professional principles and best practices
- https://ccecc.acm.org/guidance/software-engineering

Ethics

- Professional software engineers have a responsibility to society and their work carries significant liabilities
- Consequently, software engineers must conduct themselves in an ethical and professional manner
- The preamble to the Software Engineering Code of Ethics and Professional Practice [ACM 1999] states:
    - "Because of their roles in developing software systems, software engineers have significant opportunities to do good or cause harm, to enable others to do good or cause harm, or to influence others to do good or cause harm. To ensure, as much as possible, that their efforts will be used for good, software engineers must commit themselves to making software engineering a beneficial and respected profession. In accordance with that commitment, software engineers shall adhere to the following Code of Ethics and Professional Practice."

Engineering Principles

- Ability to evolve
- Efficiency
- Iteration
- Predictability
- Reliability
- Robustness
- Quality
- Security
- **Simplicity**

Software Requirements Specification (SRS)

- Software Requirements Specification (SRS) Document
- Formal document and agreement between clients and developers
- SRS is a document that describes:
  - what the software will do and how it will be expected to perform
  - the functionality the product needs to fulfill the needs of all stakeholders (business, users)

Formal Specifications (1) Purpose and High-Level Description

- Application's (product's) purpose
  - Intended Audience and Intended Use
  - Product Scope
    - Goals, objectives, benefits
  - Definitions
    - Industry-specific terms and acronyms
- Description of application
  - User needs (see Needs assessment)
  - Assumptions and Dependencies

Formal Specifications (2) - Detailed description of requirements

- Functional Requirements
- External Interface Requirements
  - User
  - Hardware
  - Software
  - Communications
- System Features
  - Functional requirements
- Non-functional Requirements
  - Availability
  - Fault tolerance
  - Maintainability
  - Performance
  - Safety
  - Security
  - Scalability
  - Sustainability
  - Usability… and more

Functional Requirements Example

- The system should have log in
- The system should store the user preferences
- The system should send a confirmation email when a user places an order
- The system should display the products' details including cost, specifications (size, color, country of origin, etc.), images (minimum 3) and a video
- The system should allow the user to add and remove items to/from a shopping bag
- The system should allow the user to pay with a variety of payment methods
- …

Functional and Non-Functional Requirements

- https://www.youtube.com/watch?app=desktop&v=9_JQUMhTzUw

Technical Feasibility

- Formal process of assessing whether it is technically possible to manufacture a product or service
    - Is it possible to develop/manufacture the application/product?
    - Does the organization have the necessary technology?
    - Does the organization have the necessary resources/personnel?
    - Can the application/product be developed with the existing technology?
    - Is the available technology the right choice?

Technical Feasibility Study Report

- Conducting a feasibility study
- A technical feasibility document is a report to manager and contains answers to the following questions:
  - Can the project plan be implemented?
  - Do the potential rewards outbalance the risks? Would be enough return on investment?

https://asana.com/resources/feasibility-study

Technical Feasibility Study Report for the Current Project

- Initial feasibility plan general for the team that includes:
  - A general high-level design report (what strategies/solutions did you searched) per team
  - A detailed individual feasibility report from each student
- The template for the student's report is:
  - What did I learned from related work and/or tutorials/ChatGPT?
  - Analysis (Positives and Negatives)
  - What solution do I propose?
  - How do I implement this solution in my portion of the project?
  - Sources that you used in your feasibility document (citations, ChatGPT).